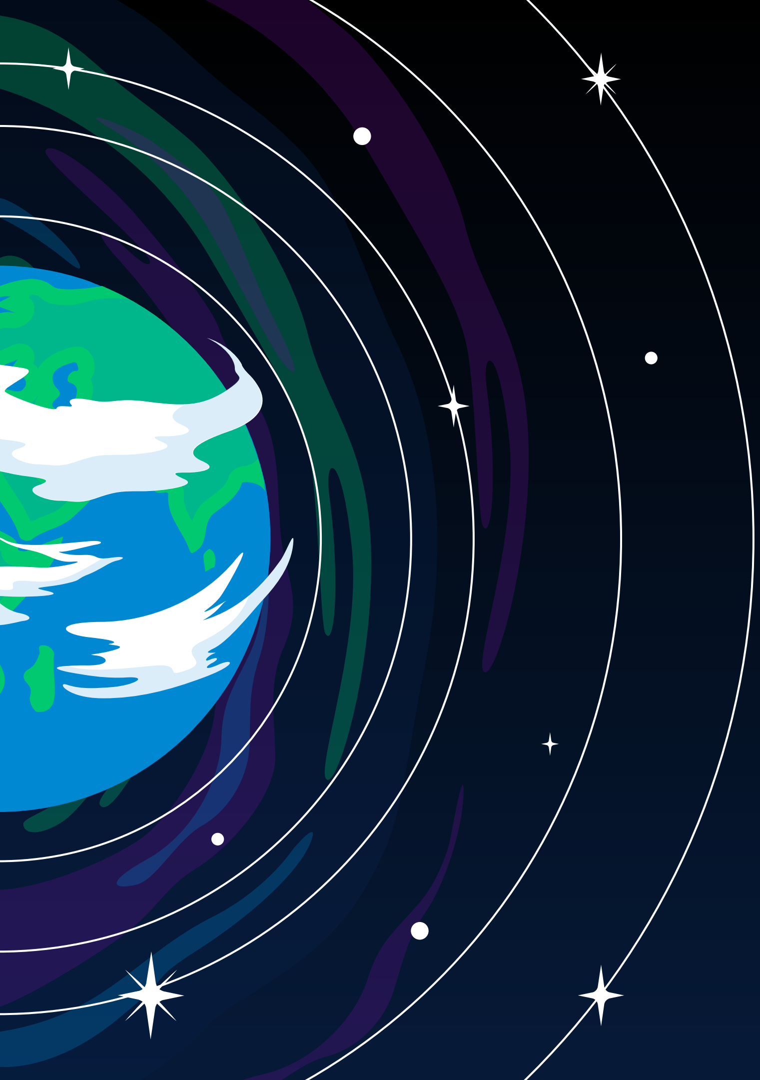


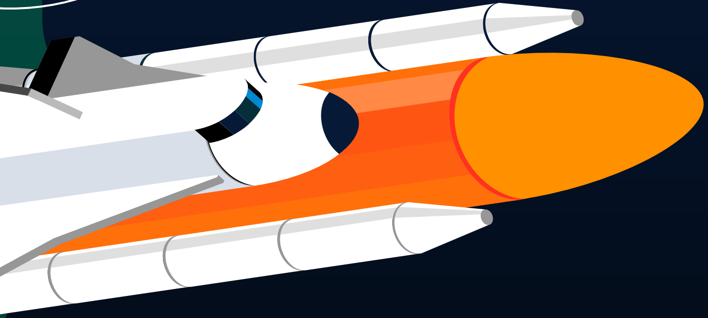
DAST

Data Across Space & Time

Gabriele Daga - Università degli Studi di Bologna
Corso di Laurea Magistrale in Ingegneria Informatica

TABLE OF CONTENT

INTRODUCTION



Satellites play a fundamental role in communication, navigation, and Earth observation. However, the increasing complexity and inter-connectivity of these systems have made them susceptible to attacks that can compromise their performance and longevity...

Goal of the project: create an anomaly detector that can spot data poisoning attacks on satellites

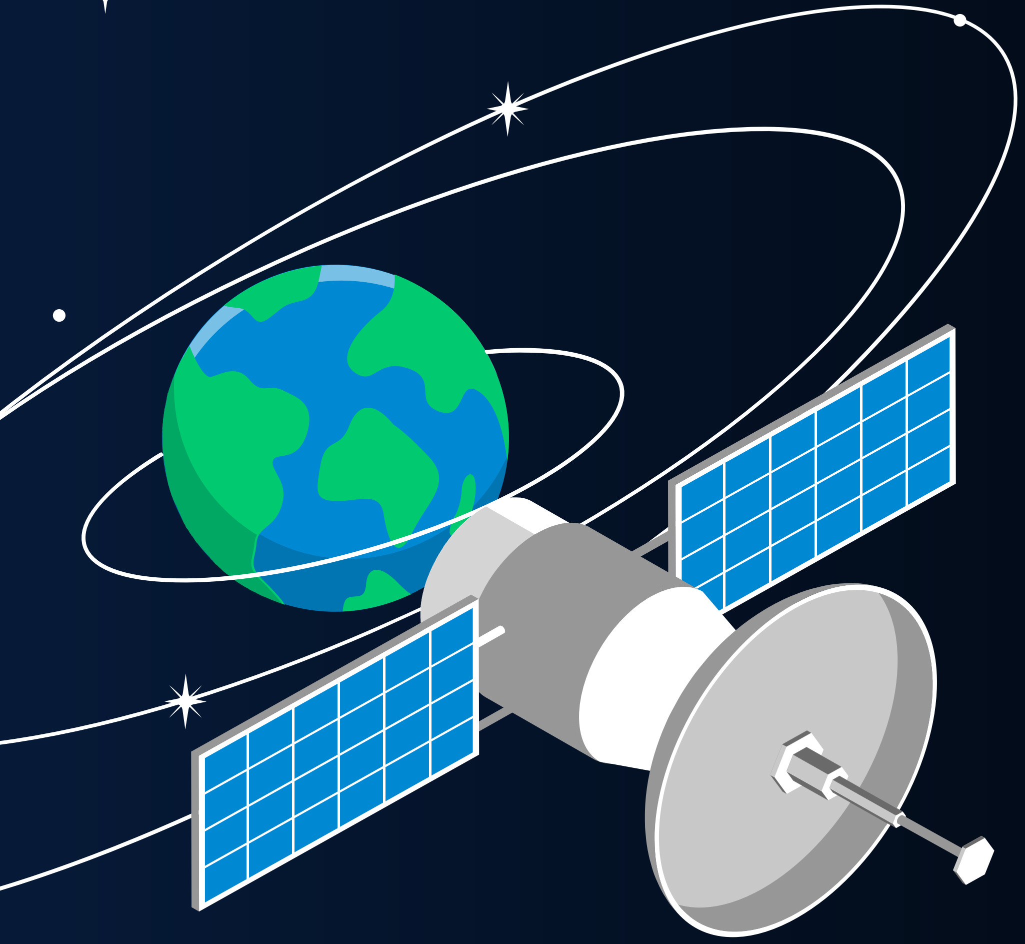
POTENTIAL ATTACKS ON SATELLITES

SENSOR SPOOFING

JAMMING AND DENIAL OF SERVICE

ADVERSARIAL ATTACKS ON MACHINE LEARNING MODELS

DATA TAMPERING

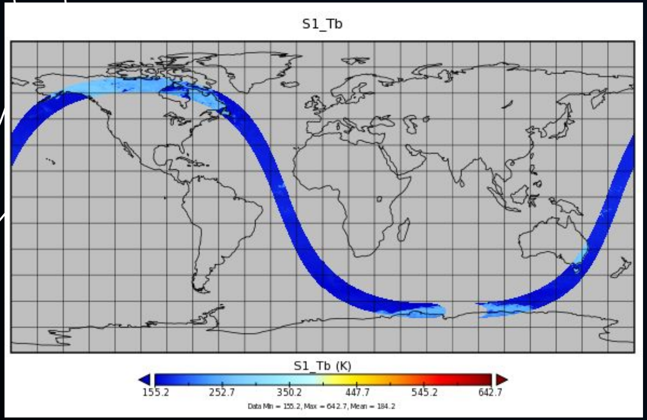


THE DATASET

SELECTED DATASETS

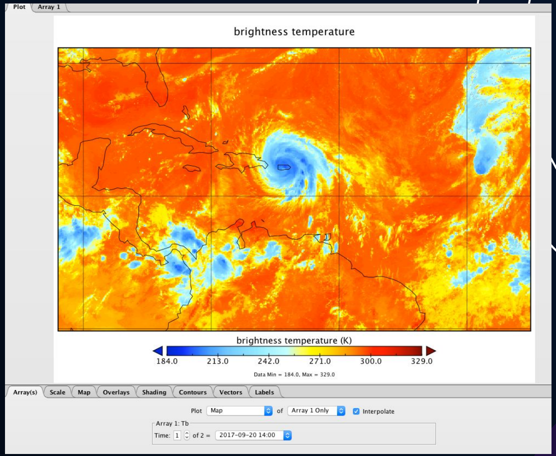
GPM (IMERG) LEVEL 1B

Employs a dual-frequency radar system known as Dual-Frequency Precipitation Radar (DPR)



NCEP/CPC L3 Half Hourly

Covers latitudes from 60°S to 60°N with a high spatial resolution of 4 km per pixel



NCEP/CPC L3 Half Hourly

Selected a subset of the dataset (otherwise too big to process)

Dimensions: (**time**: 5708, **lat**: 3298, **lon**: 9896)

Coordinates:

lon	(lon)	float32	-180.0 -179.9 ... 179.9 180.0
lat	(lat)	float32	-59.98 -59.95 ... 59.95 59.98
time	(time)	datetime64[ns]	2020-01-01 ... 2020-04-28T21:30:...

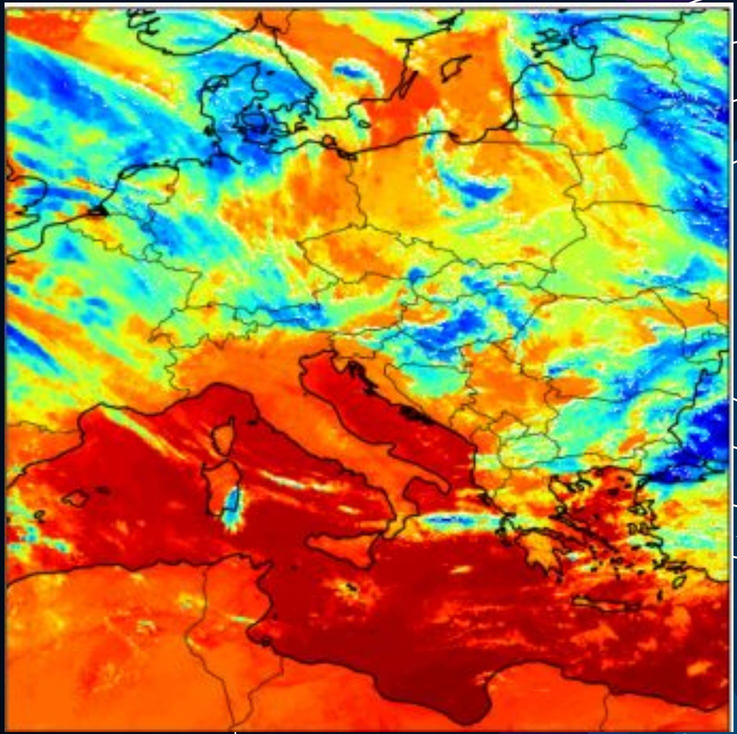
Data variables:

Tb	(time, lat, lon)	float32	dask.array<chunksize=(1, 1649, 2474), met...

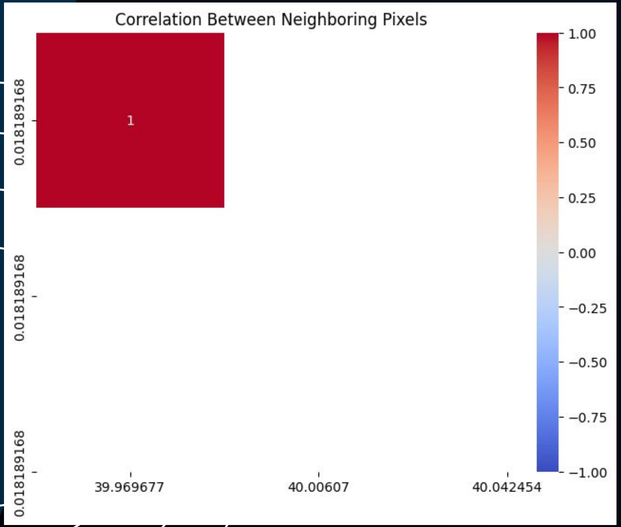
Indexes: (3)

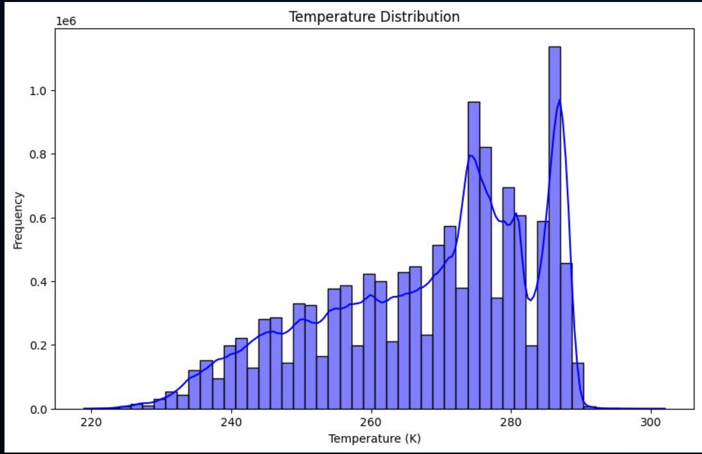
Attributes:

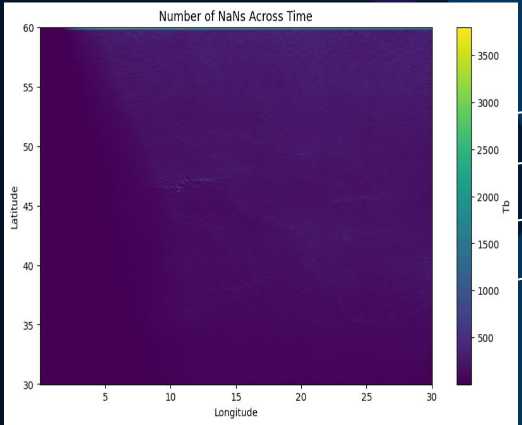
BeginDate :	2020-01-01
BeginTime :	00:00:00.000Z
EndDate :	2020-01-01
EndTime :	00:59:59.999Z
FileHeader :	StartGranuleDateTime=2020-01-01T00:00:00.000Z; StopGranuleDateTime=2020-01-01T00:59:59.999Z
InputPointer :	merg_2020010100_4km-pixel
title :	NCEP/CPC 4km Global (60N - 60S) IR Dataset
ProductionTime :	2020-01-03T00:19:23.203Z



NCEP/CPC L3 Half Hourly

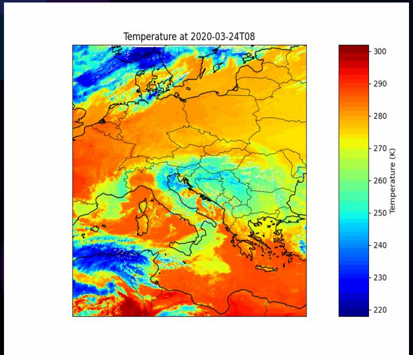






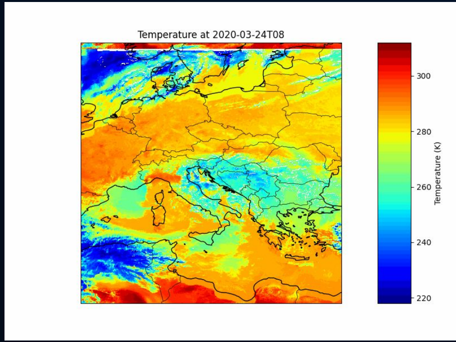
1. High spatial correlation!
2. High concentration of NaNs

TARGETED POTENTIAL ATTACKS



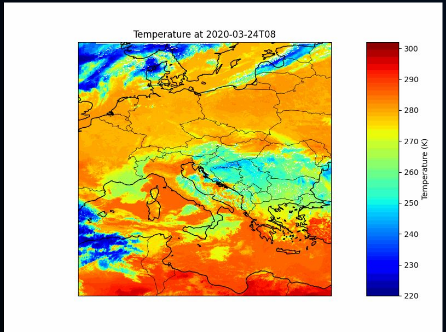
BACKDOOR ATTACK

$\tilde{x} = x + 1S \cdot z$



SENSOR SPOOFING

$xi = (xif\ , \text{if } i \in S,$
$xi, \text{otherwise.})$

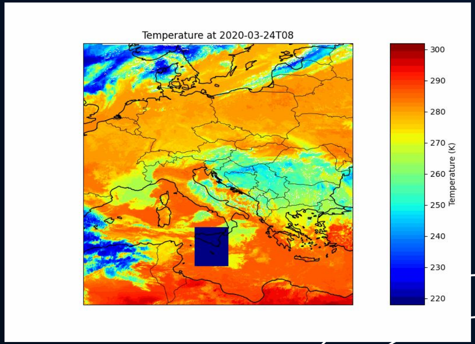


TEMPORAL SHIFTING

$x(t) = x(\lambda t + \tau)$

SPATIAL SHIFTING

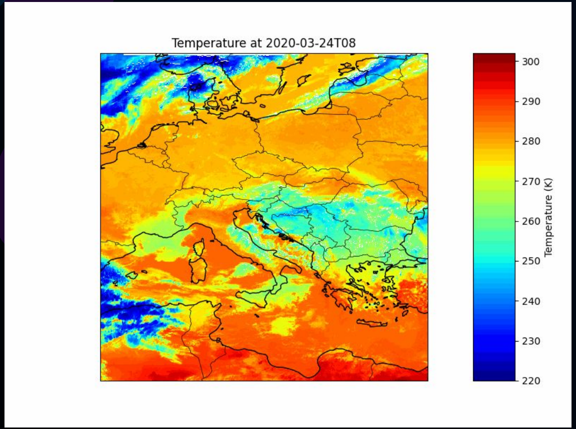
$x(p) = x(p + \Delta p)$



INDISCRIMINATE DATA POISONING ATTACKS

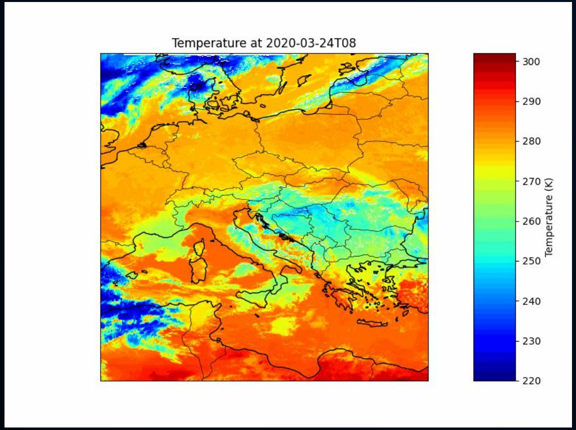
BIAS ADDITION

$$x(t) = x(t) + \alpha\, t$$



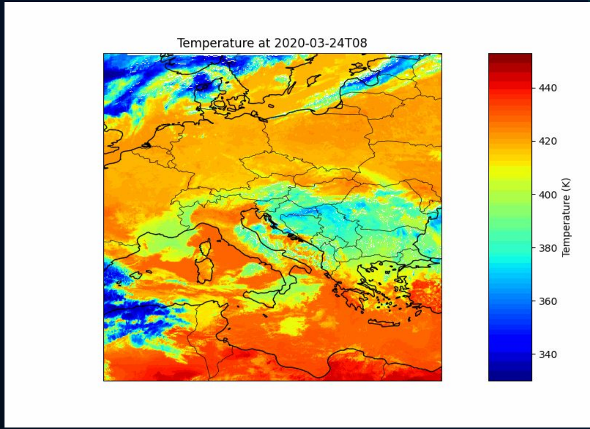
BACKDOOR ATTACK

$$\tilde{x} = x + 1S \cdot z$$



MISCALIBRATION ERROR INJECTION

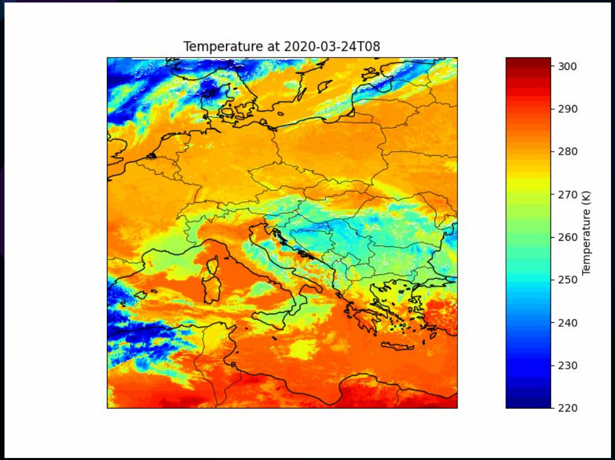
$$x = \alpha\, x$$



INDISCRIMINATE DATA POISONING ATTACKS

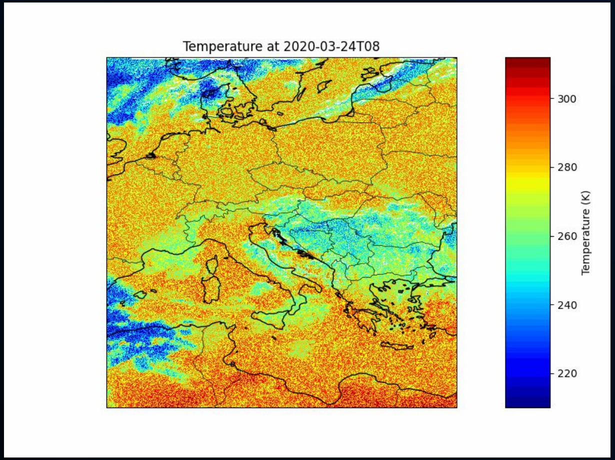
DATA BLOCK JAMMING

$xi = (NaN, \text{if } i \in S, xi, \text{otherwise})$



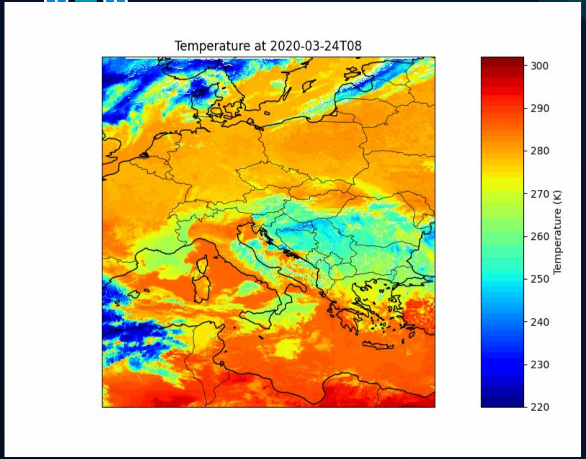
ADVERSARIAL PERTURBATION

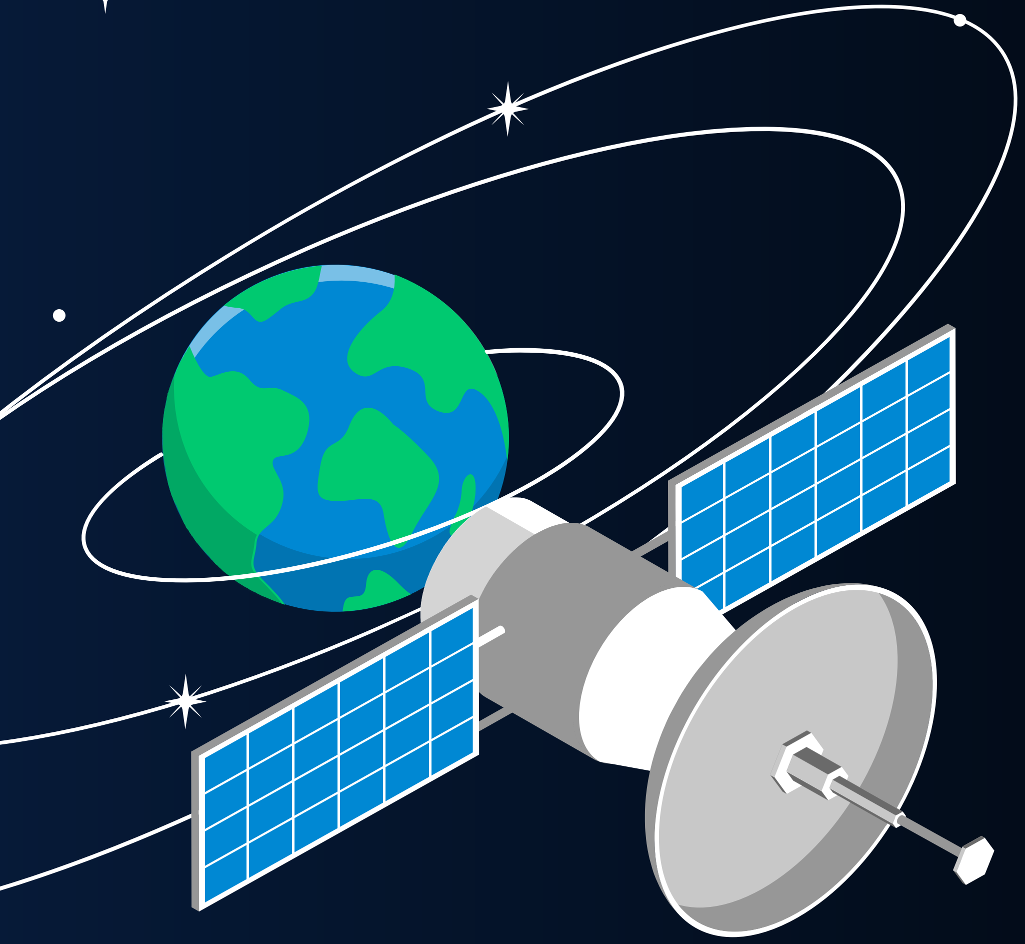
$x = x + \epsilon\text{sign}(\nabla xL(x, y))$



REPLAY ATTACK

$x(t) = x(t0)$ for $t \in [t1, t2]$

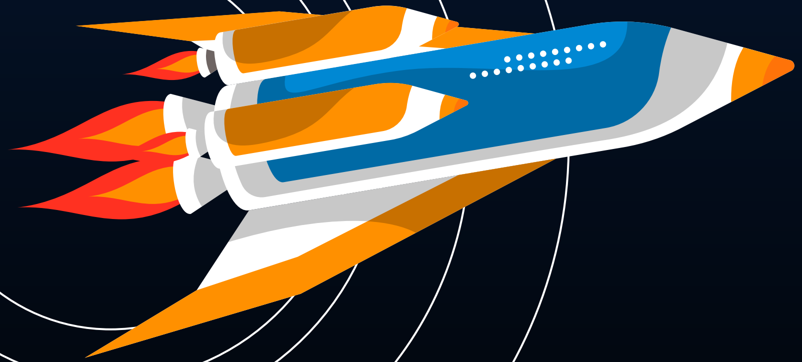


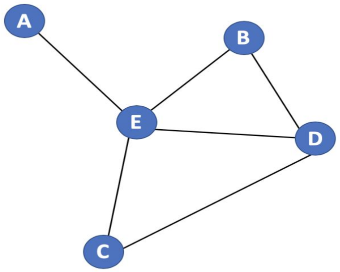


THE ANOMALY DETECTOR

Graph Neural Networks (GNNs)

Graph Neural Networs (GNNs)





Graph G

	A	B	C	D	E
A	0	0	0	0	1
B	0	0	0	1	1
C	0	0	0	1	1
D	0	1	1	0	1
E	1	1	1	1	0

Adjacency matrix A

	A	B	C	D	E
A	1	0	0	0	0
B	0	2	0	0	0
C	0	0	2	0	0
D	0	0	0	3	0
E	0	0	0	0	4

Degree matrix D

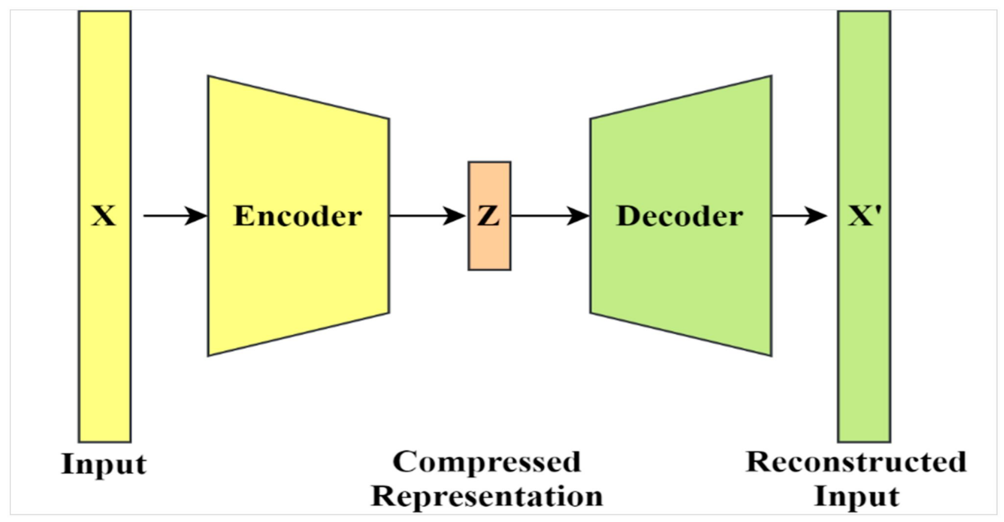
A	-1.1	3.2	4.2
B	0.4	5.1	-1.2
C	1.2	1.3	2.1
D	1.4	-1.2	2.5
E	1.4	2.5	4.5

Feature vector X

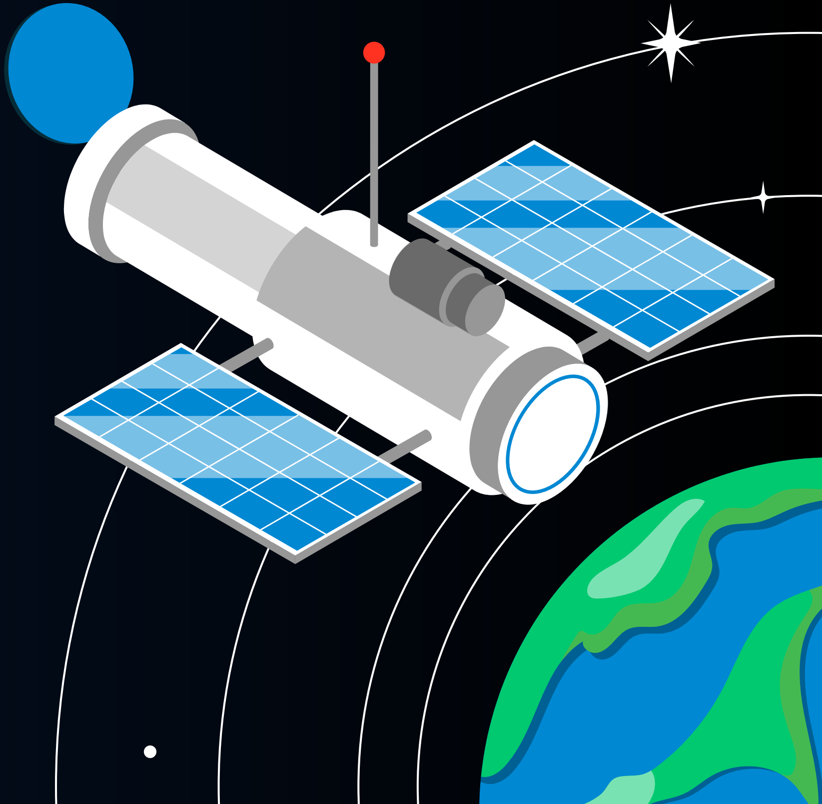
$$h_i^{(k+1)} = \mathrm{UPDATE}^{(k)}\left(h_i^{(k)}, \mathrm{AGGREGATE}^{(k)}\left(\{h_j^{(k)} : j \in N(i)\}\right)\right)$$

GNNs are well-suited for tasks where relationships and dependencies are key —such as in social networks, recommendation systems, and knowledge graph

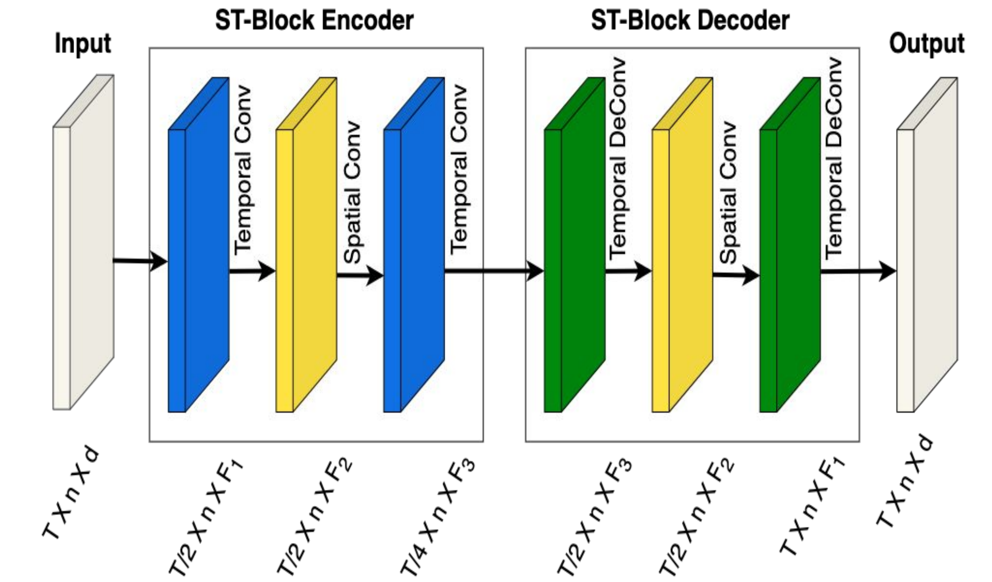
Graph Auto Encoders (GAEs)



Unsupervised technique that learns patterns from the input data and creates a lower-dimensional representation of the nodes, capturing both the feature and structural information of the input graph.



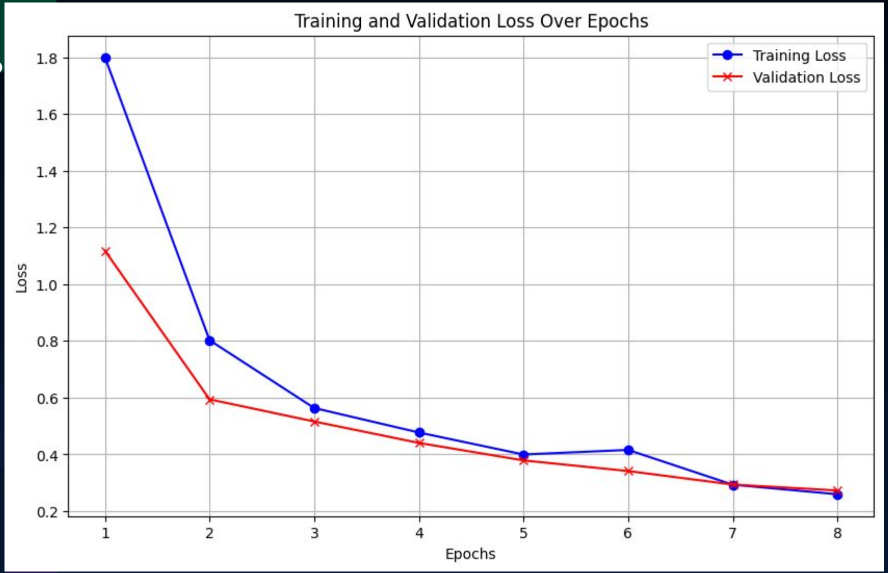
Spatio-Temporal Graph Auto Encoders (STGAEs)



T: Number of Time Points; n: Number of PV Inverters; d: Input and Output channel; F_1, F_2, F_3: Filters. Note: $d = F_1 = 1$.

STGAEs extend the concept of auto encoders — neural networks that compress data into a lower-dimensional latent space and then reconstruct the original input—by operating directly on graph-structured data.

STGAE Training



- The dataset comprises null data: every graph node will be modeled as a tuple composed by the temperature value and a mask for presence of null value
- Values normalization
- Due to the graph's large size and high memory demands, I was unable to construct extensive node neighborhoods and, as a result, had to increase the number of convolutional layers to capture relationships between farther nodes.
4. TopKPooling adoption for limited available memory constraints and in order to reduce redundancy in node connections

–

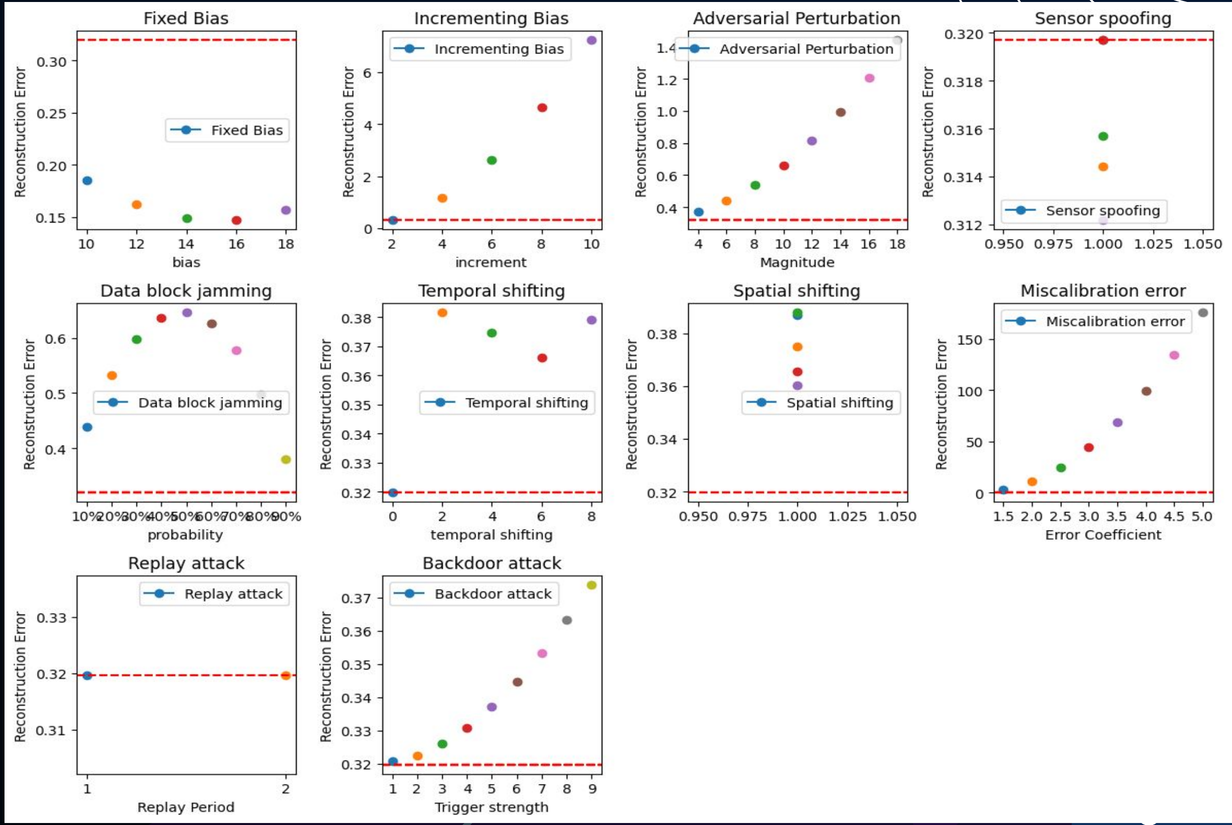
77% overall precision

Often detected:

- Backdoor attacks
- Miscalibration simulation
- Increasing bias

Often undetected:

- Constant bias
- Satellite spoofing
- Temporal shifting



Future work

- Incorporate features like sparsity, which may facilitate more interconnected neighborhoods.
- Automatic poisoning attacks with Generative Adversarial Networks (GANs)
- Employ STGAEs to measure correlations between satellite swaths